Substituting $x = a$ gives $A_k = \dfrac{f^{(k)}(a)}{k!}$ which completes the proof of the theorem.

In order to expand a function around a point say $x = a$, equivalently in powers of $(x - a)$ we need to differentiate the given function as many times as the required powers and evaluate at $x = a$. This will give the value for the coefficients of the required powers of $(x - a)$.

Example 7.30

Expand $\log(1 + x)$ as a Maclaurin's series upto 4 non-zero terms for $-1 < x \leq 1$.

Solution

Let $f(x) = \log(1 + x)$, then the Maclaurin series of $f(x)$ is $f(x) = \sum_{n=0}^{n=\infty} a_n x^n$, where, $a_n = \dfrac{f^{(n)}(0)}{n!}$ various derivatives of the function $f(x)$ evaluated at $x = 0$ are given below:

Function and its derivatives	**$\log(1 + x)$ and its derivatives**	**value at $x = 0$**
$f(x)$	$\log(1 + x)$	0
$f'(x)$	$\dfrac{1}{1 + x}$	1
$f''(x)$	$-\dfrac{1}{(1 + x)^2}$	-1
$f'''(x)$	$\dfrac{2}{(1 + x)^3}$	2
$f^{(iv)}(x)$	$-\dfrac{6}{(1 + x)^4}$	-6

Table 7.2

Substituting the values and on simplification we get the required expansion of the function given by,

$$\log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \quad ; -1 < x \leq 1.$$

Example 7.31

Expand $\tan x$ in ascending powers of x upto 5th power for $-\dfrac{\pi}{2} < x < \dfrac{\pi}{2}$.

Solution

Let $f(x) = \tan x$. Then the Mclaurin series of $f(x)$ is

$$f(x) = \sum_{n=0}^{n=\infty} a_n x^n, \text{ where, } a_n = \frac{f^{(n)}(0)}{n!}.$$

Various derivatives of the function $f(x)$ evaluated at $x = 0$ are given below:

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